

Software Defined Networking (SDN) for AI

Programmable networks for programmable intelligence



SDN Fundamentals

Architecture Split

Centralized **control plane** making decisions while distributed **data plane** handles packet forwarding

Southbound Protocols

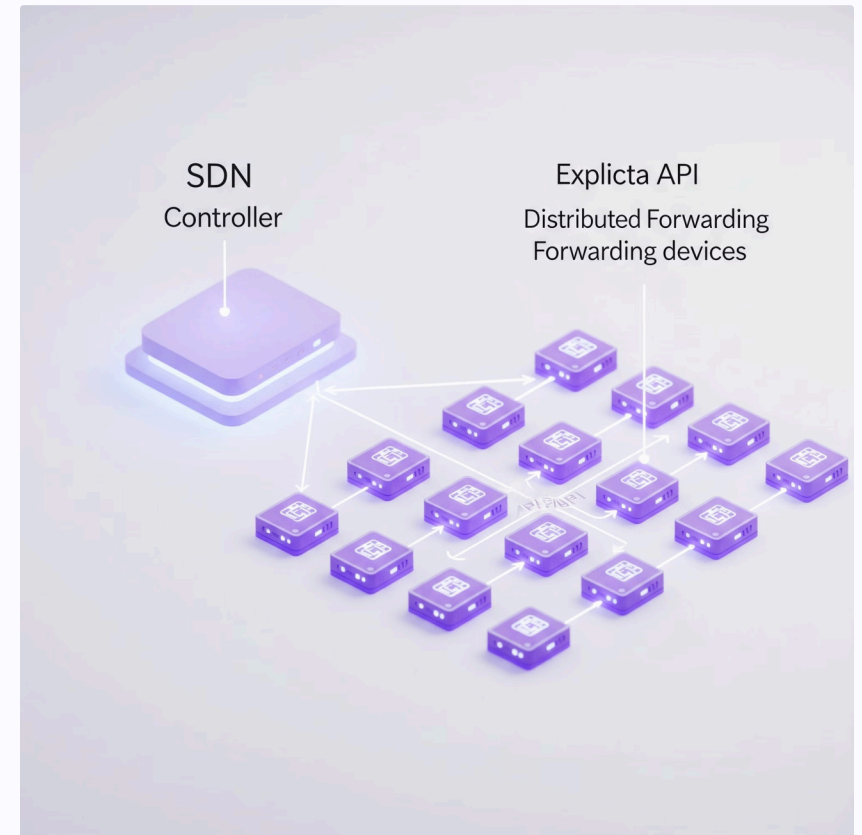
OpenFlow, P4Runtime, gNMI/gNOI enabling controller-to-device communication

Network Structure

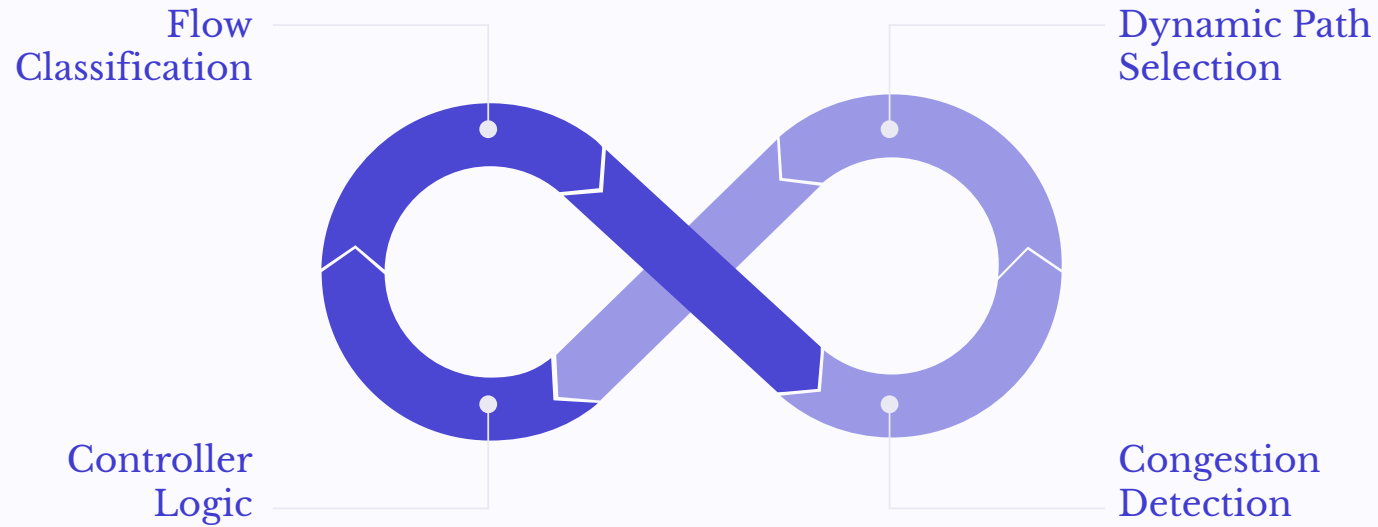
Overlays using **VXLAN/EVPN** with L3/ECMP fabric underlay for scalability

Policy as Code

Network configurations become versioned, testable, and auditable code artifacts



AI-Aware Traffic Engineering



AI workloads generate distinct traffic patterns that require specialized handling:

Flow Intelligence

- Distinguish high-volume **"elephant" flows** (all-reduce operations) from frequent but small **"mice" flows** (RPCs)
- Apply appropriate routing strategies based on flow classification

Optimization Goal

- Maximize **bisection bandwidth** for collective operations
- Dynamically adjust path selection using ECMP biasing or SRv6
- Leverage congestion signals (ECN/DCQCN) to inform controller decisions

QoS & Class Isolation

Training

High bandwidth, moderate latency sensitivity, long-lived flows

Inference

Low bandwidth, extreme latency sensitivity, bursty patterns

Storage

High bandwidth, medium latency, sequential access patterns

Control

Low bandwidth, high reliability, consistent delivery needs

Advanced implementation considerations:

- Lossless transport for RoCEv2 with carefully tuned Priority Flow Control (PFC)
- Prevent head-of-line blocking through intelligent buffer management policies
- Guarantee bandwidth allocation with priority queuing mechanisms



Multi-Tenant Segmentation

Modern AI infrastructure requires strong isolation between teams and workloads:



Tenant Isolation

Separate VRFs and VXLAN VNIs create complete logical separation between teams



Intent Definition

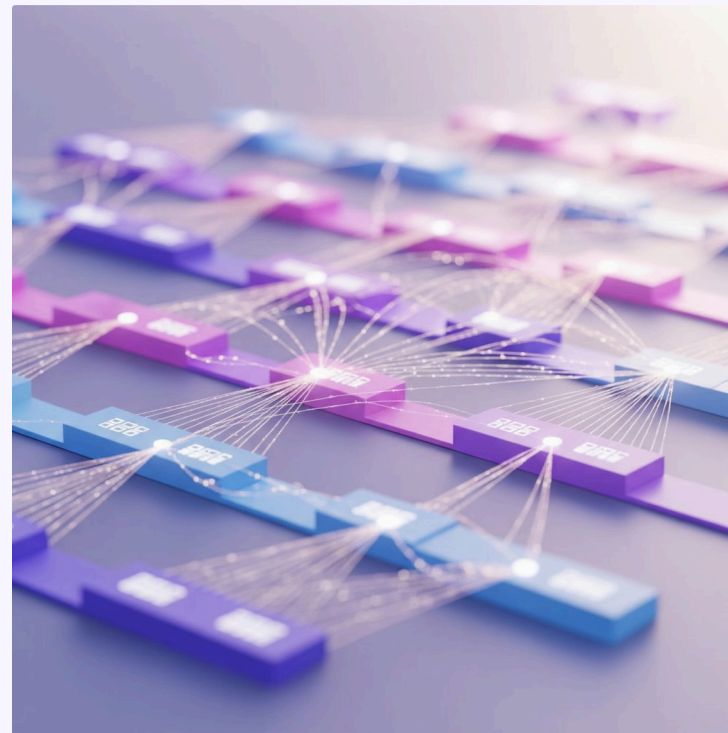
Declarative policies like "Team A can reach Feature Store; nothing else"



Micro-segmentation

Fine-grained ACLs/eBPF controls for east-west traffic within tenants

This approach enables accurate cost allocation and role-based access control (RBAC) aligned with organizational structure.



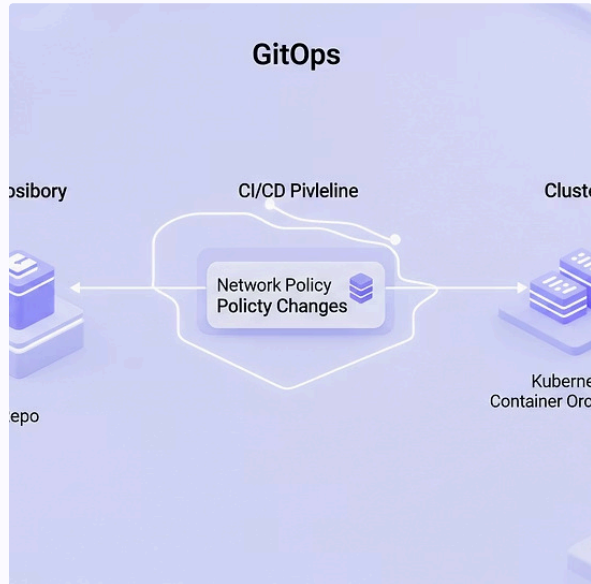
Kubernetes & SDN



CNI Options

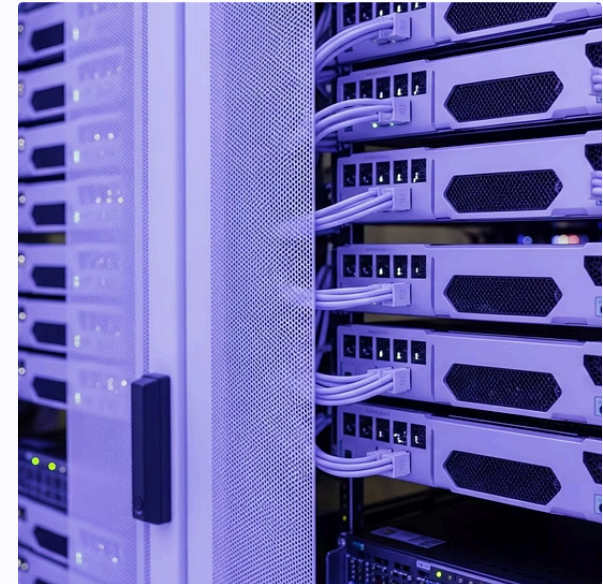
Cilium (eBPF), Calico, and Flannel provide different approaches to container networking with varying performance and security tradeoffs.

Optional service mesh implementation provides L7 policy enforcement and detailed telemetry for application-level traffic.



GitOps Deployment

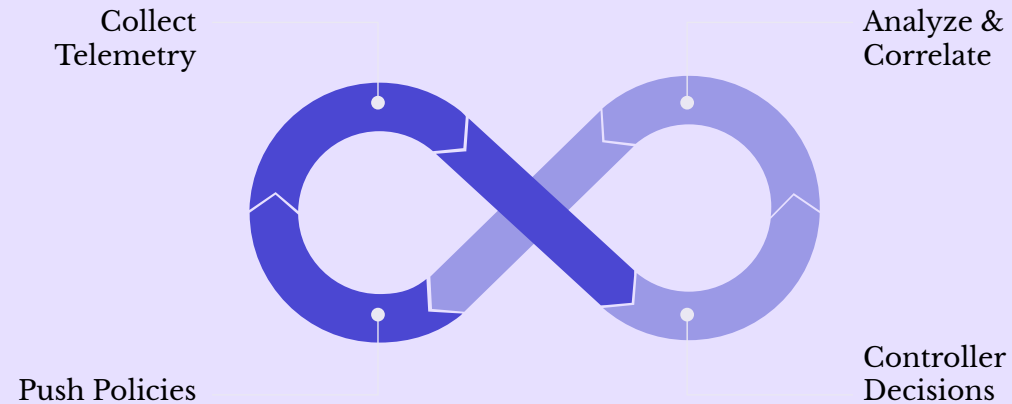
Network policies defined as code with automated testing and deployment pipelines for reliable, auditable rollouts.



Topology Awareness

Node/pod affinity configurations ensure GPU/NIC locality for optimal performance of AI workloads.

Telemetry & Closed-Loop Control



Data Collection Methods

- In-band Network Telemetry (INT)
- sFlow/NetFlow/IPFIX sampling
- gNMI streaming telemetry

Key Performance Indicators

- Flow completion time
- p99 latency measurements
- Queue depth monitoring

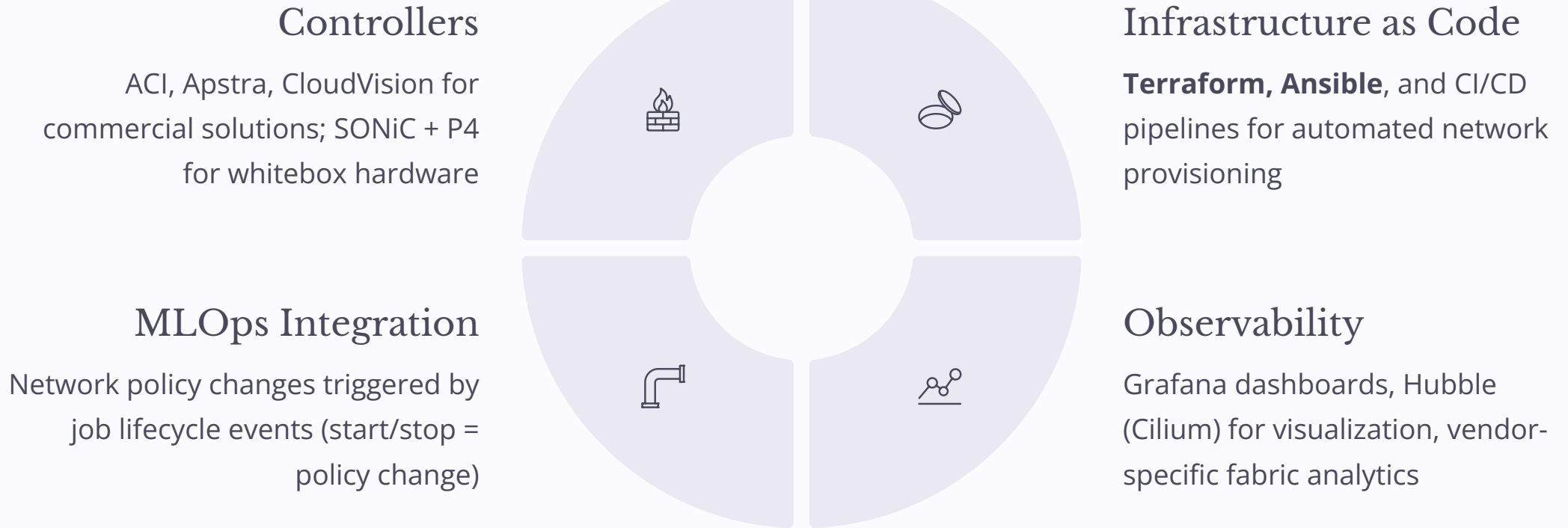
Controller Actions

- Dynamic traffic re-routing
- Bandwidth re-shaping
- Priority re-adjustment

Automation Framework

- SLO-driven policy enforcement
- Alert generation for threshold violations
- Automated remediation actions

Tooling & Ecosystem



The most effective SDN implementations leverage a combination of these tools to create a cohesive ecosystem that meets both infrastructure and application requirements.

Security with SDN

Zero-Trust Architecture

Default deny policies with explicit permissions for legitimate flows based on least-privilege principles

Encryption Capabilities

MACsec/IPsec implementation configurable per traffic class or tenant for data protection

Automated Protections

Rate limiting, DDoS shields, and other guardrails automatically enforced by the controller

Change Management

Defined change windows and continuous drift detection integrated into deployment pipeline



Pitfalls & Anti-Patterns



Controller Resilience

Avoid controller as Single Point of Failure (SPOF)

- Deploy in high-availability configuration
- Regularly test failover scenarios
- Ensure graceful degradation paths



Policy Management

Prevent uncontrolled policy sprawl

- Standardize templates and patterns
- Implement automated policy linting
- Regular policy review and cleanup



Observability Gaps

Eliminate blind spots in monitoring

- Unify underlay and overlay visibility
- End-to-end path tracing capabilities
- Correlated cross-layer analytics



Brownfield Integration

Manage transitions carefully

- Plan phased cutovers with rollback options
- Validate configurations in lab environments
- Maintain hybrid operations capabilities

Best Practices & Takeaways

Building AI-Ready Networks

Architectural Simplicity

Keep underlay network simple and stable while moving intelligence and policy logic to SDN overlays

Policy as Code

Treat network policies as code with proper reviews, testing, and version control for reliable changes

Closed-Loop Operations

Implement full telemetry → controller → action feedback loops for continuous optimization

Business Outcomes

Deliver predictable, efficient, and secure AI fabrics that scale with your computational needs



"Networks that adapt to AI workloads—not the other way around."